```
#Course: COSC 545
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```

Download Robots Data

First, download a bunch of robots.txt files.

Load in urls.

```
#from https://tranco-list.eu/list/G66YK/1000000
import csv
import os
import tldextract
tranco_rows = csv.DictReader(open("top-1m.csv"),fieldnames=["rank","domain"])#not working, need to use CSV lib

total_accepted = 0
max_accepted = 100000# you will need to have at least 10k robots.txt files
target_domains = ["us","ca","mx","br","ar","uk","ie","fr","es","pt","de","it","nl","be","ch","at","se","no","fi","dk","pl","cz","hu
accepted_domains = []
for trow in tranco_rows:
#     print(trow)
    domain = trow["domain"]
    extract = tldextract.extract(domain)
    tld = extract.suffix
#     print(tld)
    if tld in target_domains:
        accepted_domains.append(trow["domain"])
        total_accepted += 1
    if total_accepted >= max_accepted:
        break
print(accepted_domains[0:15])
print(f"Total number of domains: {len(accepted_domains)}")
```

```
['youtu.be', 'mail.ru', 'zoom.us', 'yandex.ru', 'europa.eu', 'nic.ru', 'sberdevices.ru', 'mangosip.ru', 'stbid.ru', 'rambler.ru', '
Total number of domains: 100000
```

EDA on the CSV

```
import pandas as pd


tranco_df = pd.read_csv("top-1m.csv", header=None, names=["rank", "domain"])
tranco_df.head()
```

	rank	domain
0	1	google.com
1	2	amazonaws.com
2	3	facebook.com
3	4	a-msedge.net
4	5	microsoft.com

```
tot_domains = len(tranco_df)

unique_domains = tranco_df["domain"].nunique()

print(f"Total domains in file: {tot_domains}")
print(f"Unique domains: {unique_domains}")
```

```
Total domains in file: 1000000
Unique domains: 1000000
```

Download robots.txt files

```
import urllib.request

#create folder if not present
folder_name = 'robot-texts'

# Check if the folder exists, if not, create it
if not os.path.exists(folder_name):
    os.makedirs(folder_name)

for domain in accepted_domains:
    url = "https://"+domain+"/robots.txt"
    try:
        # Open the URL
        with urllib.request.urlopen(url) as response:
            # Check if the response status code is 200 (OK)
            if response.getcode() == 200:
                # Read the raw text data
                raw_text = response.read().decode('utf-8')  # Assuming the text encoding is UTF-8
                # Output the raw text into a file
                # print(raw_text)
                outfile = open(folder_name+"/"+domain,"w")
                outfile.write(raw_text)
                outfile.close()
            else:
                print("Failed to download the file. Response code:", response.getcode())
    except Exception as e:
        print("An error occurred:",domain, e)
```

```
An error occurred: berkeley.edu HTTP Error 403: Forbidden
An error occurred: umich.edu HTTP Error 403: Forbidden
An error occurred: columbia.edu HTTP Error 403: Forbidden
An error occurred: psu.edu HTTP Error 404: Not Found
An error occurred: uci.edu HTTP Error 403: Forbidden
An error occurred: csmc.edu <urlopen error [SSL: CERTIFICATE_VERIFY_FAILED] certificate verify failed: unable to get local issuer
An error occurred: uchicago.edu HTTP Error 403: Forbidden
---------------------------------------------------------------------------
KeyboardInterrupt                         Traceback (most recent call last)
Cell In[3], line 15
     12 url = "https://"+domain+"/robots.txt"
     13 try:
     14     # Open the URL
---> 15     with urllib.request.urlopen(url) as response:
     16         # Check if the response status code is 200 (OK)
     17         if response.getcode() == 200:
     18             # Read the raw text data
     19             raw_text = response.read().decode('utf-8')  # Assuming the text encoding is UTF-8

File /usr/lib/python3.12/urllib/request.py:215, in urlopen(url, data, timeout, cafile, capath, cadefault, context)
    213 else:
    214     opener = _opener
--> 215 return opener.open(url, data, timeout)

File /usr/lib/python3.12/urllib/request.py:515, in OpenerDirector.open(self, fullurl, data, timeout)
    512     req = meth(req)
    514 sys.audit('urllib.Request', req.full_url, req.data, req.headers, req.get_method())
--> 515 response = self._open(req, data)
    517 # post-process response
    518 meth_name = protocol+"_response"

File /usr/lib/python3.12/urllib/request.py:532, in OpenerDirector._open(self, req, data)
    529     return result
    531 protocol = req.type
--> 532 result = self._call_chain(self.handle_open, protocol, protocol +
    533                           '_open', req)
    534 if result:
    535     return result

File /usr/lib/python3.12/urllib/request.py:492, in OpenerDirector._call_chain(self, chain, kind, meth_name, *args)
    490 for handler in handlers:
    491     func = getattr(handler, meth_name)
--> 492     result = func(*args)
    493     if result is not None:
    494         return result
```

```
import os
import urllib.request
from concurrent.futures import ThreadPoolExecutor, as_completed
# Faster version
folder_name = 'Robot-texts'
if not os.path.exists(folder_name):
    os.makedirs(folder_name)

def download_robots(domain):
    url = "https://"+domain+"/robots.txt"
    try:
        with urllib.request.urlopen(url, timeout=10) as response:
            if response.getcode() == 200:
                raw_text = response.read().decode('utf-8')  # Assuming the text encoding is UTF-8
                # Output the raw text into a file
                # print(raw_text)
                outfile = open(folder_name+"/"+domain,"w")
                outfile.write(raw_text)
                outfile.close()
                return f"{domain} downloaded."
            else:
                return f"{domain} failed with code {response.getcode()}."
    except Exception as e:
        return f"{domain} error: {e}"

with ThreadPoolExecutor(max_workers=35) as executor:
    futures = [executor.submit(download_robots, domain) for domain in accepted_domains]
    for future in as_completed(futures):
        print(future.result())
```

```
telecid.ru error: <urlopen error [SSL: CERTIFICATE_VERIFY_FAILED] certificate verify failed: Hostname mismatch, certificate is no
dzenmira.ru error: <urlopen error [Errno -5] No address associated with hostname>
cmedianub.ru error: <urlopen error [Errno -5] No address associated with hostname>
zoom.us downloaded.
```

```
    1090 del self._buffer[:]
    1091 self.send(msg)
    1093 if message_body is not None:
    1094
    1095     # create a consistent interface to message_body
```

```
            if hasattr(msg, 'read'):
                # Let file-like take precedence over byte-like.  This
                # is needed to allow the current position of mmap'ed
                # files to be taken into account.
File /usr/lib/python3.12/http/client.py:1035, in HTTPConnection.send(self, data)
   1033 if self.sock is None:
   1034     if self.auto_open:
-> 1035         self.connect()
   1036     else:
   1037         raise NotConnected()
File /usr/lib/python3.12/http/client.py:1470, in HTTPSConnection.connect(self)
   1467 def connect(self):
   1468     "Connect to a host on a given (SSL) port."
-> 1470     super().connect()
   1472 if self._tunnel_host:
   1473     server_hostname = self._tunnel_host
File /usr/lib/python3.12/http/client.py:1001, in HTTPConnection.connect(self)
    999 """Connect to the host and port specified in __init__."""
   1000 sys.audit("http.client.connect", self, self.host, self.port)
-> 1001 self.sock = self._create_connection(
   1002     (self.host,self.port), self.timeout, self.source_address)
   1003 # Might fail in OSs that don't implement TCP_NODELAY
   1004 try:
File /usr/lib/python3.12/socket.py:828, in create_connection(address, timeout, source_address, all_errors)
    826 host, port = address
    827 exceptions = []
-> 828 for res in getaddrinfo(host, port, 0, SOCK_STREAM):
    829     af, socktype, proto, canonname, sa = res
    830     sock = None
File /usr/lib/python3.12/socket.py:963, in getaddrinfo(host, port, family, type, proto, flags)
    960 # We override this function since we want to translate the numeric family
    961 # and socket type values to enum constants.
    962 addrlist = []
-> 963 for res in _socket.getaddrinfo(host, port, family, type, proto, flags):
    964     af, socktype, proto, canonname, sa = res
    965     addrlist.append((_intenum_converter(af, AddressFamily),
    966                      _intenum_converter(socktype, SocketKind),
    967                      proto, canonname, sa))
KeyboardInterrupt:
```

```
folder_name = "Robot-texts"


records = []
for filename in os.listdir(folder_name):
    filepath = os.path.join(folder_name, filename)
    if os.path.isfile(filepath):  # skip directories
        with open(filepath, "r", encoding="utf-8", errors="ignore") as f:
            text = f.read()
        domain = filename.replace(".txt", "")
        records.append({
            "domain": domain,
            "robots_text": text
        })
robots_df = pd.DataFrame(records)
robots_df.head()
```

domain	robots_text

```
import re

def parse_robots(text):
    user_agents = re.findall(r"(?i)User-agent:\s*([^\s#]+)", text)
    disallows = re.findall(r"(?i)Disallow:\s*([^\s#]*)", text)
    allows = re.findall(r"(?i)Allow:\s*([^\s#]*)", text)
    return user_agents, disallows, allows

robots_df["user_agents"] = robots_df["robots_text"].apply(lambda x: parse_robots(x)[0])
robots_df["disallows"] = robots_df["robots_text"].apply(lambda x: parse_robots(x)[1])
robots_df["allows"] = robots_df["robots_text"].apply(lambda x: parse_robots(x)[2])

robots_df.head()
```

	domain	robots_text	user_agents	disallows	allows
0	merca2.es	User-agent: *\nDisallow: /search/\nDisallow: /...	[*]	[/search/, /*?s=, /?s=, /wp-admin/]	[/search/, /*?s=, /?s=, /wp-admin/]
1	bloknot-voronezh.ru	User-Agent: *\nCrawl-delay: 20\nDisallow: /sea...	[*]	[/search/, /bitrix/, /admin/, /upload/, /img/,...	[/search/, /bitrix/, /admin/, /upload/, /img/,...
2	homestoreandmore.ie	User-agent: *\n\n# Exclude search from indexin...	[*]	[/*Search-Show*, /*Search-GetSuggestions*, /se...	[/*Search-Show*, /*Search-GetSuggestions*, /se...

EDSA

```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt

print(len(robots_df))
```

```
10519
```

Finding the most frequent bots

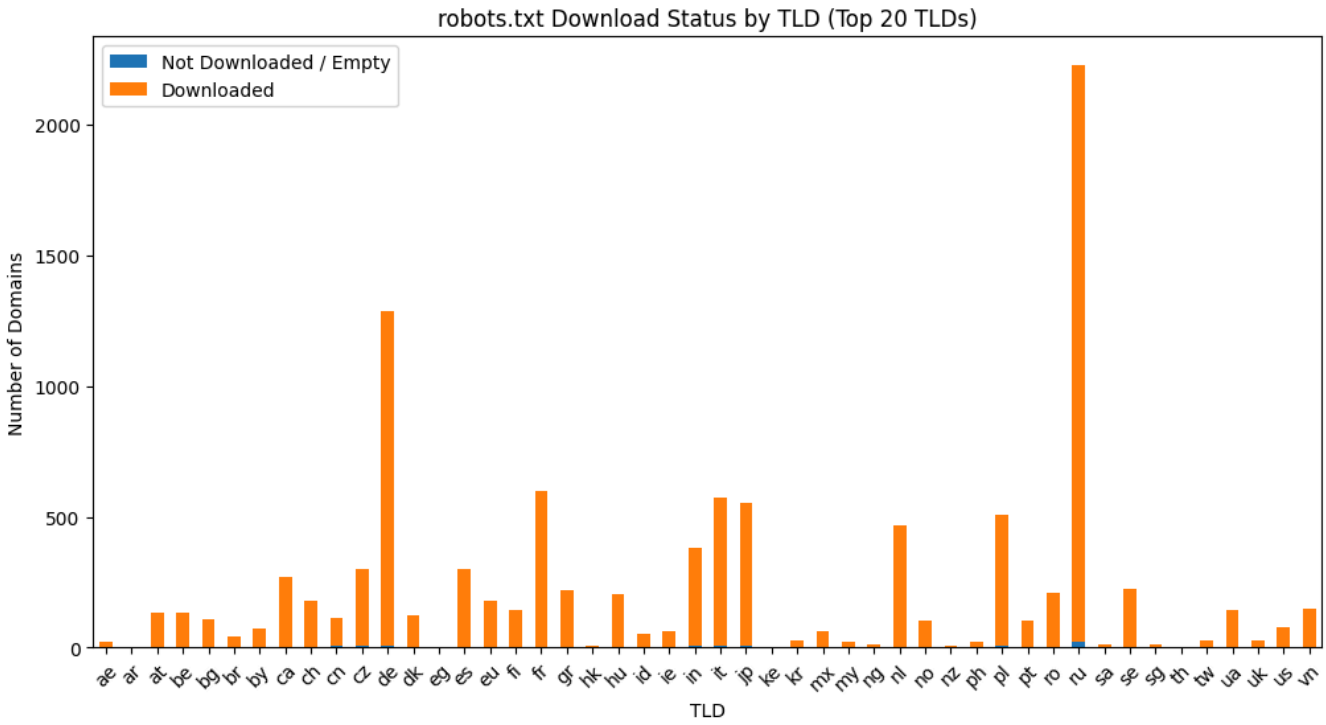
```
agents_df = robots_df[["domain", "user_agents"]].explode("user_agents")
```

```
frequent_agents = agents_df["user_agents"].value_counts()
frequent_agents.head(10)
```

```
user_agents
*                9938
Yandex           1281
GPTBot           1249
CCBot            1095
Googlebot         937
ClaudeBot         829
Bytespider        820
ChatGPT-User      806
Google-Extended   801
AhrefsBot         798
Name: count, dtype: int64
```

```
import matplotlib.pyplot as plt
import pandas as pd
import tldextract


extractor = tldextract.TLDExtract()
robots_df["tld"] = robots_df["domain"].apply(lambda d: extractor(d).suffix)

# was robots.txt successfully read (non-empty)?
robots_df["downloaded"] = robots_df["robots_text"].apply(lambda x: bool(x.strip()))

tld_status = robots_df.groupby("tld")["downloaded"].value_counts().unstack(fill_value=0)

tld_status.plot(kind="bar", stacked=True, figsize=(12,6))
```

```
plt.title("robots.txt Download Status by TLD (Top 20 TLDs)")
plt.ylabel("Number of Domains")
plt.xlabel("TLD")
plt.xticks(rotation=45)
plt.legend(["Not Downloaded / Empty", "Downloaded"])
plt.show()
```



Most Blocked Agents

```
def get_blocked_agents(text):
    lines = text.splitlines()
    blocked = []
    current_agent = None
    disallow = False

    for line in lines:
        line = line.strip()
        if line.lower().startswith("user-agent:"):
            if current_agent and disallow:
                blocked.append(current_agent)
            current_agent = line.split(":", 1)[1].strip()
            disallow = False
        elif line.lower().startswith("disallow:"):
            path = line.split(":", 1)[1].strip()
            if path:  # non-empty path = blocked
                disallow = True
            if current_agent and disallow:
                blocked.append(current_agent)
            return blocked
```

```
robots_df["blocked_agents"] = robots_df["robots_text"].apply(get_blocked_agents)

blocked_df = robots_df[["domain","blocked_agents"]].explode("blocked_agents")
blocked_counts = blocked_df["blocked_agents"].value_counts()
```

```
print(blocked_counts)
```

```
blocked_agents
*                               6613
Yandex                           336
Googlebot                        118
Amazonbot                        111
```

```
GPTBot                                103
                                     ...
*           \t# Todos los crawlers      1
MSIECrawler                             1
RyteBotSprd_DEV-209115                  1
TosCrawler                              1
WBSearchBot                             1
Name: count, Length: 240, dtype: int64
```

Domain Comparison: .us versus .eu

```
us_eu_df = robots_df[robots_df["tld"].isin(["us", "eu"])]

print(us_eu_df)
```

```
10374       datadoghq.eu  User-agent: *\nAllow: /sb/\nAllow: /s/\nAllow:...
10458  bandainamcoent.eu  #\n# robots.txt\n#\n# This file is to prevent ...
10498      chaturbate.eu  <!DOCTYPE html PUBLIC "-//W3C//DTD XHTML 1.0 S...
10502          blebox.eu         User-agent: *\n\nAllow: /\n\nDisallow: /*?

                                          user_agents  \
114                                              [*]
219                                               []
230                                               []
298                                              [*]
360                                              [*]
...                                              ...
10364                                             []
10374                                            [*]
10458  [*, ChatGPT-User, GPTBot, CCBot, Google-Extend...
10498                                             []
10502                                            [*]

                                            disallows  \
114                                                 []
219                                                 []
230                                                 []
298                          [/login.php, /register.php]
360                                                [/]
...                                                ...
10364                                               []
10374                                              [/]
10458  [/core/, /profiles/, /README.txt, /web.config,...
10498                                               []
10502                                           [/*?]

                                               allows tld  downloaded  \
114                                                []  eu        True
219                                                []  us        True
230                                                []  us        True
298    [/index.php, /media.php, /login.php, /register...  eu        True
360                                               [/]  eu        True
...                                               ...  ..         ...
10364                                              []  eu        True
10374                                 [/sb/, /s/, /$, /]  eu        True
10458  [/core/*.css$, /core/*.css?, /core/*.js$, /cor...  eu        True
10498                                              []  eu        True
10502                                       [/, /*?]  eu        True

      blocked_agents
114             None
219             None
230             None
298              [*]
360              [*]
...              ...
10364           None
10374            [*]
10458            [*]
10498           None
10502            [*]

[256 rows x 8 columns]
```

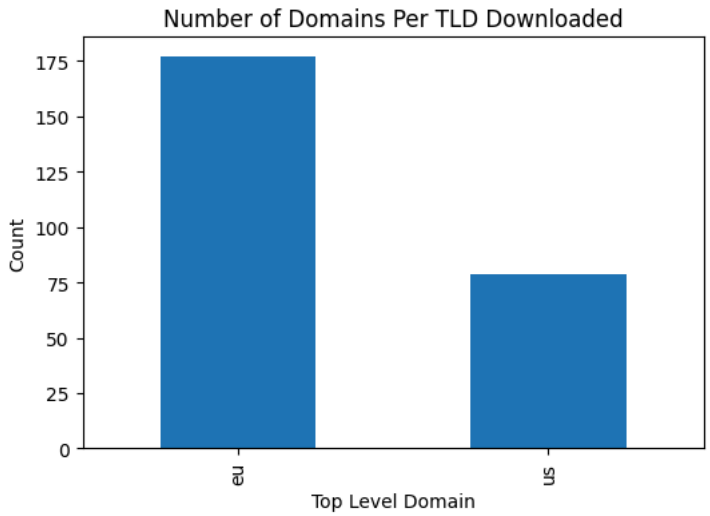
Number of successful Robots.txt downloads

```
us_eu_count = us_eu_df.groupby("tld")["robots_text"].apply(lambda x: x.str.strip().ne("").sum())
print(us_eu_count)
```

```
tld
eu    177
us     79
Name: robots_text, dtype: int64
```

```
plt.figure(figsize=(6,4))
us_eu_count.plot(kind="bar")
plt.title("Number of Domains Per TLD Downloaded")
plt.xlabel("Top Level Domain")
plt.ylabel("Count")
```

```
Text(0, 0.5, 'Count')
```



Access vs. Restriction

```
us_eu_df["fully_blocked"] = robots_df["disallows"].apply(lambda x: "/" in x)
fully_blocked = us_eu_df.groupby("tld")["fully_blocked"].count()
print(fully_blocked)
```

```
tld
eu    177
us     79
Name: fully_blocked, dtype: int64
/tmp/ipykernel_112807/2506923988.py:1: SettingWithCopyWarning:
A value is trying to be set on a copy of a slice from a DataFrame.
Try using .loc[row_indexer,col_indexer] = value instead

See the caveats in the documentation: https://pandas.pydata.org/pandas-docs/stable/user_guide/indexing.html#returning-a-view-versus
  us_eu_df["fully_blocked"] = robots_df["disallows"].apply(lambda x: "/" in x)
```

AI Bot access

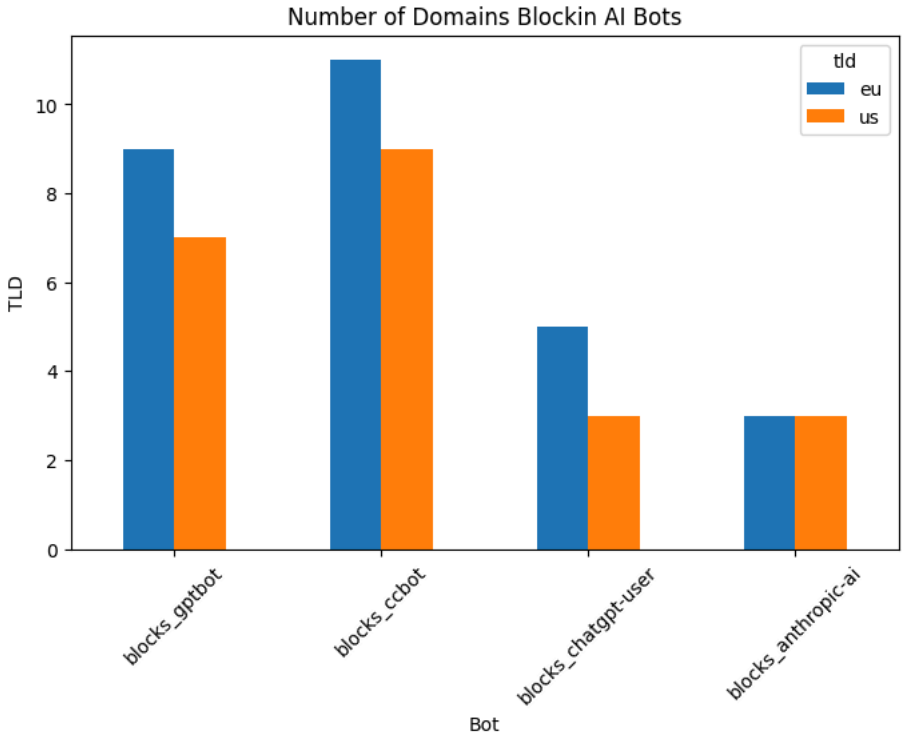
```
ai_bots = ["GPTBot", "CCBot", "ChatGPT-User", "anthropic-ai"]
for bot in ai_bots:
    us_eu_df[f"blocks_{bot.lower()}"] = us_eu_df["robots_text"].str.contains(bot, case=False, na=False)
ai_blocked = {bot: us_eu_df[f"blocks_{bot.lower()}"].sum() for bot in ai_bots}
ai_df = pd.DataFrame(list(ai_blocked.items()), columns=["Bot", "Count"])

print(ai_df)
```

```
           Bot  Count
0       GPTBot     16
1        CCBot     20
2  ChatGPT-User      8
3  anthropic-ai      6
/tmp/ipykernel_112807/474322461.py:3: SettingWithCopyWarning:
A value is trying to be set on a copy of a slice from a DataFrame.
Try using .loc[row_indexer,col_indexer] = value instead

See the caveats in the documentation: https://pandas.pydata.org/pandas-docs/stable/user_guide/indexing.html#returning-a-view-versus
  us_eu_df[f"blocks_{bot.lower()}"] = us_eu_df["robots_text"].str.contains(bot, case=False, na=False)
```

```
ai_count_tld = us_eu_df.groupby("tld")[[f"blocks_{bot.lower()}" for bot in ai_bots]].sum().T

ai_count_tld
```

tld	eu	us
blocks_gptbot	9	7
blocks_ccbot	11	9
blocks_chatgpt-user	5	3
blocks_anthropic-ai	3	3

```
ai_count_tld.plot(kind="bar", figsize=(8,5))
plt.title("Number of Domains Blockin AI Bots")
plt.xlabel("Bot")
plt.ylabel("TLD")
plt.xticks(rotation=45)
plt.show()
```



Reflection Section

Q : What type of selection did you do for your urls and how many robots.txt files did you collect?

Answer: I ultimately filtered down urls by country. There was no set process for choosing which countries, I just googled "tld's for countries" and gathered a couple random tld's. Originally I started with academia, but there were so many refused connections, I switched to something more broad to get more robot.txt files. In total, I grabbed 10,348 files.

Q Which user-agent(s) are blocked the most in your dataset? Do you have an idea of why?

Answer: Most files just blocked all user-agents, but of the named ones, YandexBot was the most blocked. Yandex is a russian search engine that employs YandexBot to crawl web pages and other search engines to